Source Localization Using a Single Array of Sensors in Nearfield Configuration, and Application to Active Noise Cancellation

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***Abstract*—In this paper we demonstrate by computed examples that an array of sensors in nearfield configuration can provide source localization in both x and y, going beyond the usual directional localization provided by a 1-dimensional array. This ability to localize a source opens new application areas. An application presented here is the possibility of doing active noise cancellation of the sound of a snorer.**

I. Introduction

The basic concept of array processing is that signals at different sensors are delayed to add coherently, with the amount of delay at each sensor a function of the distance from source to sensor. The vast majority of array processing algorithms employ a far-field assumption in which arriving signals are taken to be plane waves. A rule of thumb is that a signal can be considered far-field at a distance of $r = 2L^2/\lambda$ where r is the radial distance from the array origin, L is the largest array dimension, and λ is the operating wavelength [1], [2]. Another rule of thumb is that a source at least ten wavelengths from a sensor is far-field. Under the far-field assumption, a linear array of sensors can determine only the direction to a source, but not the location of the source so that generally speaking, it would require two far-field linear arrays to determine where the direction vectors cross to establish the location of the source. This work considers the alternative case of nearfield array processing. While there has been sophisticated work on nearfield array processing [3], [4], [5], [6], [7], [8], [9], [10], [11], [1], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], the existing literature rarely provides visual, experimental/simulation demonstration of the localization capability, when this is even mentioned. This paper demonstrates by numerically computed results that for sources sufficiently close to the sensor array the location of the source can be determined.

Having established that sensor localization is possible an interesting potential application is suggested: using an array of microphones and speakers to do active cancellation of snoring. Despite the humor instilled by the concept of snoring, snoring is a widespread problem, affecting up to a quarter of the population of at any one time [28]; other studies report estimates of 25% of men and 15% of women are affected. While there are several associated impacts due to snoring, perhaps the most obvious is its impact on those around them. For example, surgical modification of snoring been shown to have a strong statistical impact on quality of marital state and quality of life [29]. The negative effects of snoring accompanying obstructive sleep apnea is reported in [30], where partners reported that snoring "interfered with their own sleep onset," or sleep maintenance. In a controlled study [31], "elimination of snoring (using a CPAP device) was associated with an improvement in the quality of the bed partners' sleep," resulting in "an additional 62 minutes of sleep per night for the spouses of the snorers" [31, At abstract].

While several treatments for snoring itself have been proposed (including surgery and various appliances such as CPAP and mouthpieces and nose fixtures), the work proposed here approaches the problem at the recipient's' end: to actively cancel the sound of snoring in a non-contact way, to dramatically reduce the sound of snoring, thereby eliminating a source of ongoing nocturnal irritation. This application thus eliminates the need for potential earplugs or earmuffs on the part of the listener, or devices and sleep posture modifiers on the part of the snorer. We consider a system employing an array of speakers directed to the ear of a listener, and an array of microphones steered to listen at the location of the ear. An adaptive cancellation system adjusts the speakers to minimize the measured sound at the location of the ear. (Alternatively, a steered laser Doppler vibrometer (LDV) as a noncontact microphone can be used in place of the microphone array.) The system is completely non-contact, so there are no uncomfortable fixtures or appliances to be worn which could potentially negatively affect sleep patterns in their own way. Other work applying signal processing techniques to snore analysis and cancellation are in [32], [33]. This preliminary work shows that the problem is of interest, but that there is

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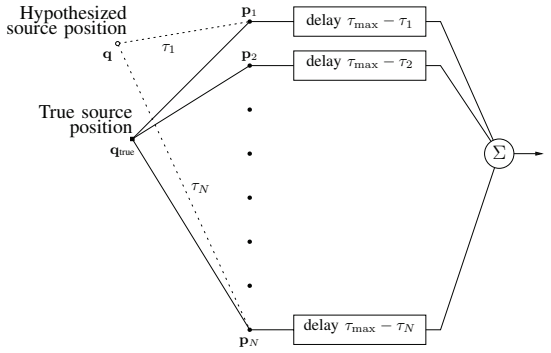


Fig. 1. Delay-and-sum array processing

still much to be done.

We have given proposed the system the name Active Noise Control Snoring Effect Reduction, or ANCSER .

II. Delay-and-Sum Combining

Figure 1 shows a delay-and-sum combiner. Sensors are at positions $\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_N$. A source at location $\mathbf{q}_{\text{true}}$ produces a signal $x(t)$ so that the sensor i measures $y_i(t) = x(t - \tau_i)$ where $\tau_i = \|\mathbf{p}_i - \mathbf{q}_{\text{true}}\|/c$, where c is the speed of propagation. If the source location were known, the sensed signals could be delayed to align the signal together

$$y_i(t - (\tau_{\max} - \tau_i))$$

then added as

$$z(t) = \sum_{i=1}^{N} y_i(t - (\tau_{\max} - \tau_i)).$$

Here, $\tau_{\max}$ is the maximum delay. These delays align the signals so that they add coherently, producing maximum signal strength.

When the source location is not known, a hypothesized position $\mathbf{q}$ is swept over a feasible region, with delays determined by the hypothesized position. When $\mathbf{q} = \mathbf{q}_{\text{true}}$ these delays will result in coherent summation. At other positions the delays will not result in coherent addition of the received signals. A map of the energy as a function of the hypothesized position provides information about the true source location.

Knowing the location of a source, an array of sensors can be steered to attend primarily to that location. Turning this around, the relative timing at different speakers in an array of speakers can be used to steer the primary response of the net acoustic field to a particular location. Both the steered speaker and steered microphone are used in the noise cancelling application described below.

III. Creating signal delays

For a narrowband signal, $s(t) = m(t)e^{j2\pi f_c t}$ the signal delayed signal

$$s(t - \tau) = \approx m(t)e^{j2\pi f_c t}e^{-j2\pi f_c \tau} = s(t)e^{-j2\pi f_c \tau},$$

so that the delay results in a phase change. We consider in this paper, however, the case that the signal is broadband so that this approximation is not valid. For a signal that is to be sampled, the delay can be represented using the Shannon-Whittaker formula [34]. The delayed signal is

$$s(t - \tau) = \sum_{\ell=-\infty}^{\infty} s(\ell T)\,\text{sinc}(2B((t - \tau) - \ell T)),$$

where $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$. The sampled delayed signal is

$$s(nT - \tau) = \sum_{\ell} s(\ell T)\,\text{sinc}(2B(nT - \ell T - \tau))$$

In practice the sum is truncated as

$$s(nT - \tau) = \sum_{\ell=-P}^{P} s(\ell T)\,\text{sinc}(2B(nT - \ell T - \tau))$$

This is recognized as a convolution of the sequence of signal samples $\{s(nT)\}$ with the impulse response

$$h(n) = \text{sinc}(2B(nT - \tau)), n = -P, ..., P$$

In practice this is stored causally in an array, which introduces a bulk delay in the signal but does not change the relative delays among the array of sampled signals.

IV. Examples

Figure 2 illustrates some examples of array response. The 9 sensors (microphones) are indicated by the white squares along $y = 0$, separated by $d = 0.25$ m to form an aperture of width 2m. The signal tested on is a male speaker. In this figure, the plot on the left shows the linear response and the plot on the right shows the response in dB. In part (a) the true source position is (0.5, 0.25). The array response clearly peaks at the true position. As the true position is moved further from the sensors, the array response elongates, so the array increasingly senses only the direction to the source and not the particular location of the source.

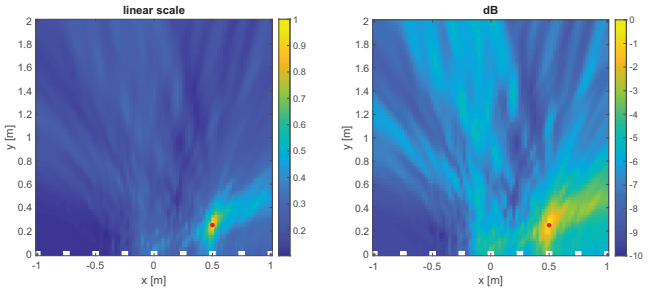
The general delay-and-sum framework does not require the sensors to be in a linear array, or evenly spaced. As another example, Figure 3 illustrated a scenario where the 9 sensors are arranged in an L shape. The delay-and-sum algorithm is capable of locating the source, though with more spread in the x direction.

V. Application: Noise cancellation of a snoring signal

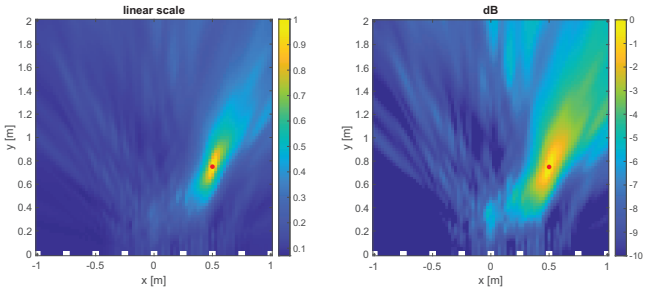
Since most array processing algorithms are far field, these results raise the question: under what circumstances would it be useful to do near-field array processing. A potential application is in the cancellation of snoring signals. Despite its humorous-sounding nature, as mentioned in the introduction snoring is a widespread phenomenon with possible negative health effects. The goal of this application is to create a way to mitigate the sound of snoring from a person sharing a bed, with minimal physical impact — no apparatus to put on.

The physical concept is illustrated in figure 4. Two people sleep in a bed equipped with a speaker array capable of steering a snore-cancelling signal from a speaker array,

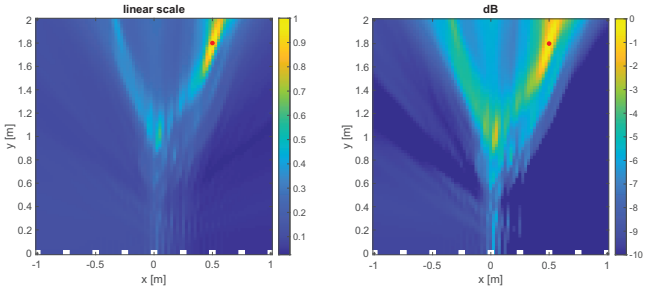
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(a) True source position = (0.5, 0.25)

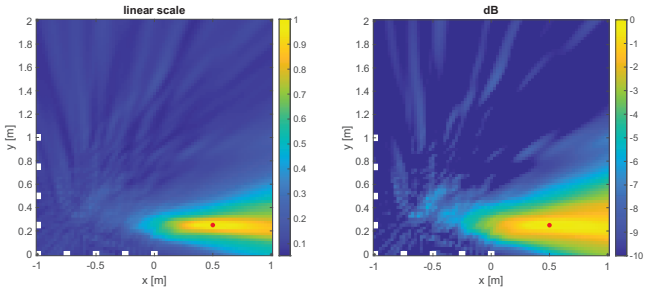


(b) True source position = (0.5, 0.75)

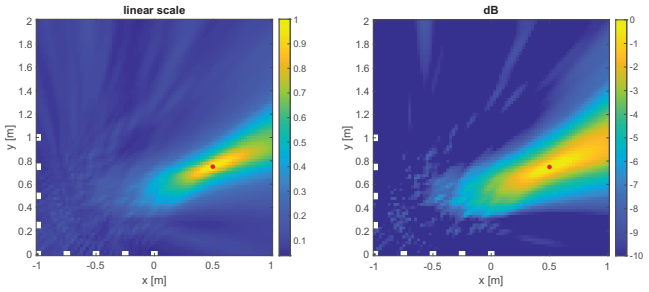


(c) True source position = (0.5, 1.8)

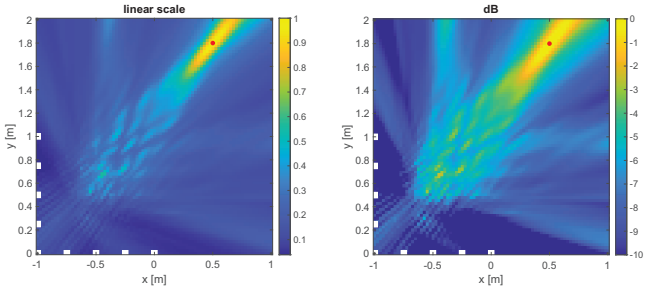
Fig. 2. Near-field delay-sum array response, linear array



(a) True source position = (0.5, 0.25)



(b) True position = (0.5, 0.75)



(c) True source position = (0.5, 1.8)

Fig. 3. Near-field delay-sum array response, L-shaped array

focussed on the ear of a listener. A camera and software provides pattern recognition and tracking to keep the beam focussed at the generally correct location. A microphone near the snoring person picks up the snoring signal, which is passed to algorithms which predicate and compensate for acoustic transfer functions and steer a cancelling signal directed to the general area of the ear of the other sleeper. The cancellation algorithm adapts the signal such that the signal measured at the ear of the sleeper, as determined by a steered microphone array, has minimum energy. (The microphone array with different arrays could also pick up the snoring sound, but a microphone placed near the snorer will reduce propagation delay, which is critical for this problem.)

Figure 5 illustrates the signal processing block diagram. The snorer produces a signal $s(t)$ which propagates through the acoustic transfer function H_{sh} to the ear of the hearer as $r(t)$. The snoring signal also propagates through the acoustic transfer function H_{sm} to the microphone, where it produces the signal $m(t)$, which is provided as an input to noise cancellation algorithm. The outputs of the noise cancellation algorithm, signals $c_1(t), \ldots, c_N(t)$, are delayed to steer their collective beam to the general location of the hearer's ear, then propagate through acoustic transfer functions H_{1h}, ..., H_{Nh} to the ear of the hearer. The microphone array is steered by its delays to listen to the general location of the hearer's ear. Signals picked up by the microphone experience acoustic transfer functions $H_{h1}, \ldots, H_{hN}$. The steering of the beams is determined by the position information, detected by the camera system. Each of the acoustic transfer functions introduces (among other things) a delay that must be accounted for. The acoustic transfer functions are not known, in general.

The system can be abstracted as follows. The net cancellation signal, which incorporates the effect of the acoustic transfer functions H_{ih} and the effect of steering, is $c(t)$, which we take to be the output of the adaptive cancellation system. The transfer function of the adaptive cancellation system is

$$G(z) = g_0 + g_1 z^{-1} + \cdots + g_{N-1} z^{-(N-1)}$$

The net error signal, which incorporates the effect of the acoustic transfer functions H_{hi} and the effect of steering, is $e(t)$. We model the error (of the sampled discrete-time signal)

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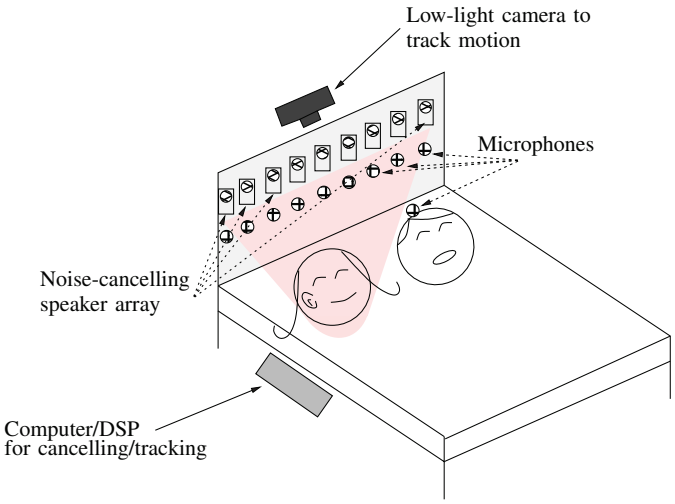


Fig. 4. Active noise cancellation of snoring signals

simply as

$$e[n] = c[n] - r[n] = \sum_{i=0}^{M-1} m[i]g[n-i] - r[n]$$
$$= \mathbf{m}[n]^T \mathbf{g}[n] - r[n],$$

where

$$\mathbf{m}[n] = \begin{bmatrix} m[n] \\ m[n-1] \\ \vdots \\ m[n-N+1] \end{bmatrix} \qquad \mathbf{g} = \begin{bmatrix} g_0 \\ g_1 \\ \vdots \\ g_{N-1} \end{bmatrix}.$$

We establish a criterion: determine the filter coefficient vector $\mathbf{g}$ to minimize $J(\mathbf{g}) = E[e[n]^2]$. Taking the gradient we find

$$\frac{\partial J(\mathbf{g})}{\partial \mathbf{g}} = 2E[(c[n] - r[n])\mathbf{m}[n]] \approx 2e[n]\mathbf{m}[n],$$

where the approximation is the stochastic gradient approximation. The filter coefficients are updated using an LMS-like, steepest descent on the stochastic gradient as

$$\mathbf{g}^{[n+1]} = \mathbf{g}^{[n]} - \mu \mathbf{m}[n]e[n]$$

The signal $m[n]$ is not known exactly at the canceller, being the snoring signal modified by the unknown acoustic transfer function H_{sm}. Nor are any of the other acoustic transfer functions H_{hi} or H_{ih} known, each of which will introduce elements of delay. However, the error signal $e(t)$ depends on these, so the expectation is that the filter will adapt to minimize the error in a way that accommodates these transfer functions.

VI. Conclusions and Discussions

As this paper has demonstrated by simulation examples, a nearfield array is able to provide position localization, not merely direction to source as is conventionally believed for array processing. As the distance of the source to the sensors increases, the region of localization elongates, ultimately providing essentially only directional information.

From these simulations, the width of the localization region across the direction of the sensors is roughly the same as the beamwidth for direction finding purposes, with the cross direction width being approximation three times that width.

The possibility of source localization with a linear array opens up new application areas. One of these is using an array of speakers and microphones in a nearfield source cancellation application, specifically for cancelling of snoring noise.

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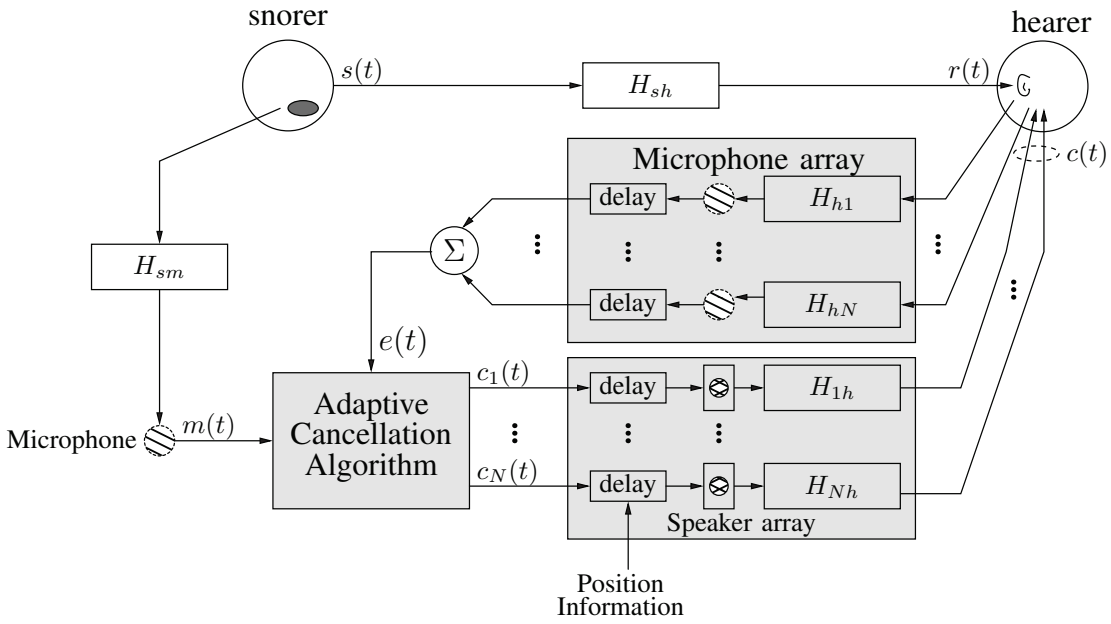


Fig. 5. Signal processing block diagram for snore cancellation

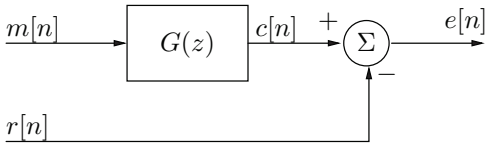


Fig. 6. Simplified block diagram of noise cancellation system

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